External Test Dataset Preparation

An external fire and smoke dataset was used to test the trained fire and smoke detection model on new data. The dataset was obtained from CQUniversity / Figshare. The dataset is titled *Annotated Fire-Smoke Image Dataset for fire detection Using YOLO*.

Dataset source link:
https://acquire.cqu.edu.au/articles/dataset/Annotated_Fire_-Smoke_Image_Dataset_for_fire_detection_Using_YOLO_/28747046

Direct download link used:
https://acquire.cqu.edu.au/ndownloader/files/53486522

The dataset contains labeled fire and smoke images in YOLO format, with bounding-box annotation files in .txt format. It originally had two classes, fire and smoke. Since the trained model uses one combined class called fire_smoke, all valid annotations were converted to class 0. This means both fire and smoke objects were treated as one class during testing.

The downloaded ZIP file was extracted and cleaned. All valid image and label pairs found in the dataset were combined into one test dataset. The final dataset was arranged in YOLO format with an images/test folder, a labels/test folder, and a data.yaml file.

During cleaning, 11,021 images were found. Out of these, 11,019 images were kept for testing. No images were missing label files. Two images were removed because their label files had no valid bounding boxes. No invalid annotation rows were skipped.

The final cleaned test dataset contained 11,019 valid image-label pairs. This dataset was used only for external testing, not for training or validation. Therefore, it helped measure how well the model performs on new unseen fire and smoke images.

External Test Dataset Preparation and Model Testing

An external fire and smoke dataset was used to test the trained fire and smoke detection model on new data. The dataset was obtained from CQUniversity / Figshare. The dataset is titled *Annotated Fire-Smoke Image Dataset for fire detection Using YOLO*.

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The model was tested using an image size of 640, a confidence threshold of 0.25, and an IoU threshold of 0.5. The model achieved a precision of 80.15%, recall of 64.47%, F1-score of 71.46%, mAP50 of 75.47%, mAP75 of 59.22%, and mAP50-95 of 53.56%.

The inference speed was 6.27 ms per image. The preprocessing time was 0.85 ms, while postprocessing took 0.17 ms. These results show that the model was able to detect fire and smoke reasonably well on unseen external data. The high precision shows that most predictions made by the model were correct. However, the lower recall shows that the model still missed some fire and smoke objects. Overall, the model showed acceptable external testing performance, but recall can still be improved in future work.